Digital Payments (UPI) and Financial Inclusion across Indian States

B.Tech Project, Part I

Devansh Gupta

Supervisor: Prof. Rudra Prakash Pradhan (VGSOM)
IIT Kharagpur

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Motivation

- UPI was launched in 2016 and is operated by NPCI.
- The Jan Dhan account drive began in 2014.
- Policy discussion assumes that digital payments widen access to formal finance and support growth.
- The evidence at state level is thin.
- It is unclear whether higher adoption leads to deeper financial inclusion, or whether states that were already more inclusive adopted faster.

Research questions and hypotheses

- RQ1: Does higher digital payment adoption raise financial inclusion in a state? (H1: yes, positively.)
- RQ2: Does adoption contribute to state-level economic growth after controlling for other drivers? (H2: yes, positively.)
- RQ3: Is the causality bidirectional? (H3: yes; inclusion also encourages adoption.)
- RQ4: Does the effect differ between better-connected and weaker-infrastructure states? (H4: stronger where internet access is higher.)

Literature and gap

- Finance and economic growth.
- Financial inclusion and ICT in Indian states [11, 10].
- Digital payments and UPI [12].
- Methods: panel Granger causality [3]; panel VAR [5, 1].
- **Gap:** no peer-reviewed state-panel study of UPI adoption, financial inclusion and growth with a causality test.

Data availability

- NPCI publishes state-wise UPI data only from April 2025 [8, 4].
- One analysis reports that 27.5% of FY25 UPI volume has no state attribution [6].
- The RBI Financial Inclusion Index is published for India only, not for states [13].
- The PMJDY state-wise statistics are one snapshot (21 October 2026) with no history [2].

PhonePe Pulse as proxy

- PhonePe Pulse commit `257bf38` [9].
- Coverage: 2018Q1 to 2026Q2, 35 state ids.
- Three variables: transaction count, amount (Rs.), registered users.
- PhonePe has about 46% of UPI volume.
- Total transactions: 139.98 million in 2018Q1; 38,664.29 million in 2026Q2.
- Supervisor approval of the proxy is an open item.

Variables and sample

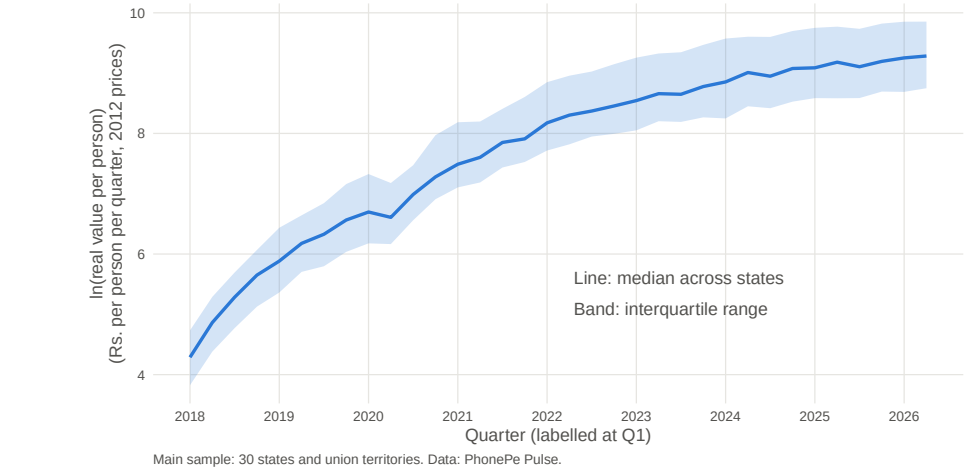
- Main sample: 30 ids (28 states, Delhi, Jammu and Kashmir including Ladakh).
- Panel: 1190 rows, 35 state ids, 34 quarters (2018Q1 to 2026Q2).
- Money values are real (state CPI, base 2012 = 100).

Variable	N	Mean	SD	Min	Max
Transactions per person (count)	1,020	8.35	13.00	0.02	100.30
Real value per person (Rs., 2012 prices)	1,020	6,130.78	8,601.89	17.48	61,997.31
Registered users per 1,000	1,020	274.04	193.56	6.99	943.50
Population (millions)	1,020	45.66	50.37	0.66	243.31
CPI	1,020	169.85	21.64	131.37	223.93

Method plan

- Step 0: description and validation (trends, maps, σ- and β-convergence, PhonePe against NPCI).
- Step 1: series properties (Pesaran CD test, CIPS or IPS, Pedroni cointegration).
- RQ1 and RQ4: fixed and random effects, Hausman test, Driscoll–Kraay standard errors.
- RQ3: Dumitrescu–Hurlin test; panel VAR by GMM.
- RQ2: annual tier, fixed effects and system GMM.
- Results are reported as predictive (Granger) causality, not as causal effects.

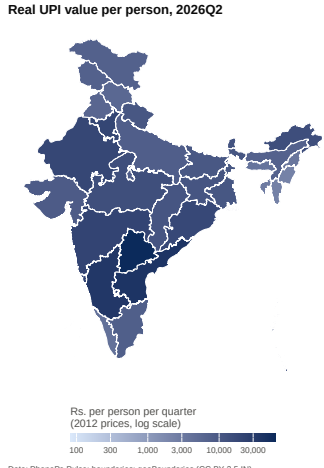
Adoption trend



Median and interquartile range across states of ln real value per person per quarter (Rs., 2012 prices), 2018Q1–2026Q2, main sample. Data: PhonePe Pulse; MoSPI state CPI; NCP 2020.

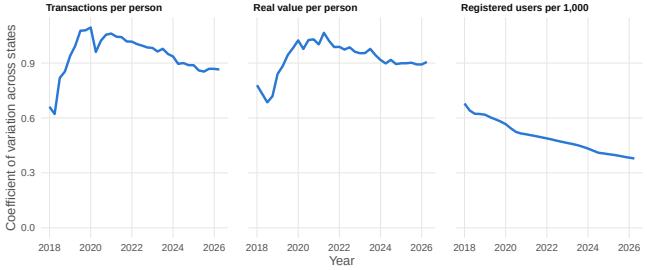
Maps: 2018Q4 and 2026Q2





Real value per person per quarter (Rs., 2012 prices); log colour scale, same scale in both panels. Left: 2018Q4. Right: 2026Q2.

Convergence



Main sample: 30 states and union territories. CV = sd / mean, per quarter. Data: PhonePe Pulse.

σ-convergence: coefficient of variation across states, 2018Q1–2026Q2.

	Real value per person	Transactions per person
Intercept	0.3773	0.5751
	(0.0788)	(0.0211)
ln(initial level)	0.0174	-0.0030
	(0.0138)	(0.0128)
N	30	30
R2	0.054	0.002

β-convergence, N = 30, OLS; standard errors in parentheses.

NPCI validation

	Pearson		Spearman	
Quarter	Count share	Value share	Count share	Value share
2025Q2	0.923	0.923	0.967	0.961
2025Q3	0.905	0.898	0.958	0.959
2025Q4	0.897	0.897	0.954	0.955
2026Q1	0.901	0.897	0.953	0.956
2026Q2	0.901	0.897	0.956	0.954

- 35 ids, 5 quarters (2025Q2 to 2026Q2).
- State shares: Pearson 0.897 to 0.923; Spearman 0.953 to 0.967.
- Per-capita levels: Pearson 0.615 to 0.742; Spearman 0.455 to 0.605.
- NPCI unclassified volume: 0.42 to 0.45 of the national total from June 2025.

Plan for BTP-II

- December to April.
- FI data (RBI BSR, PMJDY, TRAI, NSDP) and quarterly and annual FI indices (PCA; Sarma index as robustness).
- Steps 1 to 3: series properties; RQ1, RQ3, RQ4, RQ2; robustness.
- Thesis; paper (`elsarticle`) after the BTP-II results.
- District-panel extension checked after BTP-I.
- Blocking items: supervisor acceptance of the PhonePe proxy; department BTP rules.

References I

[1] Michael R. M. Abrigo and Inessa Love. Estimation of panel vector autoregression in Stata. *The Stata Journal*, 16(3):778–804, 2016. S08.

[2] Department of Financial Services. Pradhan mantri jan dhan yojana: state-wise statistics. https://pmjdy.gov.in/statewise-statistics, 2026. D04; snapshot as on 21 October 2026.

[3] Elena-Ivona Dumitrescu and Christophe Hurlin. Testing for Granger non-causality in heterogeneous panels. *Economic Modelling*, 29(4):1450–1460, 2012. S05.

[4] Chaitanya Kohli. Maharashtra leads UPI transactions as NPCI releases state-wise data. MediaNama, 14 July 2025, https://www.medianama.com/2025/07/223-maharashtra-leads-upi-transactions-npci/, 2025. N01.

[5] Inessa Love and Lea Zicchino. Financial development and dynamic investment behavior: Evidence from panel VAR. *The Quarterly Review of Economics and Finance*, 46(2):190–210, 2006. S07.

[6] Ananya Matta. Here is how UPI usage differs across Indian states. Dataful Insights, 23 January 2026, https://insights.dataful.in/articles/here-is-how-upi-usage-differs-across-indian-states, 2026. N02.

[7] National Commission on Population. Report of the technical group on population projections: India and states 2011–2036. Technical report, Ministry of Health and Family Welfare, Government of India, July 2020. URL https://nhm.gov.in/New_Updates_2018/Report_Population_Projection_2019.pdf. D05; Table 8, projected total population as on 1 March.

[8] National Payments Corporation of India. UPI ecosystem statistics: state-wise data. https://www.npci.org.in/what-we-do/upi/upi-ecosystem-statistics, 2026. D03; state-wise from April 2025; accessed 1 November 2026.

[9] PhonePe. PhonePe Pulse data repository. https://github.com/PhonePe/pulse, commit 257bf38e3095020f9f516c4c72c10bdc348781b1, 2026. D01; CDLA-Permissive-2.0; data 2018Q1–2026Q2; accessed 1 November 2026.

[10] Rudra P. Pradhan and Pradipta P. Sahoo. Financial inclusion, mobile telephony and economic growth in Indian states. *Applied Economics Letters*, 28(4):310–314, 2021. S02; verify exact title and second author's given name.

References II

[11] Rudra P. Pradhan, Mak B. Arvin, Mahendhiran Nair, et al. Economic growth, financial inclusion and ICT infrastructure development in Indian states. *Technological Forecasting and Social Change*, 169:120758, 2021. doi: 10.1016/j.techfore.2021.120758. S01; verify exact title and full author list.

[12] Aarti Reddy, Mansi Kedia, and Sanjana Shukla. Diffusion of digital payments in India: Insights based on data from PhonePe Pulse. Working Paper IPCIDE-wp1, Indian Council for Research on International Economic Relations (ICRIER). URL https://icrier.org/pdf/IPCIDE-wp1.pdf. S03; year not shown on page, verify.

[13] Reserve Bank of India. Financial inclusion index for March 2025. https://website.rbi.org.in/web/rbi/-/press-releases/financial-inclusion-index-for-march-2025, 2025. O01; press release.